

## 17-7 PH Stainless Steel, TH1050, bar and forgings

**Categories:** [Metal](#); [Ferrous Metal](#); [Stainless Steel](#); [Precipitation Hardening Stainless](#); [T S10000 Series Stainless Steel](#)

**Material** Properties are measured in the longitudinal direction.

### Notes:

**Processing:** TH1050 - heated to austenitic range, 1040°C (1900°F), and water quenched. Reheated to 760°C (1400°F) to precipitate carbides. Carbides cause the austenite transform to martensite after cooling below 15°C (60°F). Tempered at 565°C (1050°)

**Applications:** high strength high temperature applications, chemical processing equipment, heat exchangers, power boilers, superheater tubes

**Corrosion Resistance:** 17-7 PH is suitable for use in fresh water, industrial and marine atmospheres, and mild chemical and oxidizing environments. 17-7 PH should not be used in salt water or reducing environments.

**Key Words:** UNS S17700, double treatment alloy, TH1050, bar and forgings, 17-7PH, 17-7 PH, 17/7PH, 17/7 PH, Precipitation Hardening

**Vendors:** [Click here](#) to view all available suppliers for this material.

Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

| Physical Properties | Metric                    | English                                  | Comments |
|---------------------|---------------------------|------------------------------------------|----------|
| Density             | <a href="#">7.80</a> g/cc | <a href="#">0.282</a> lb/in <sup>3</sup> |          |

| Mechanical Properties      | Metric                   | English                    | Comments |
|----------------------------|--------------------------|----------------------------|----------|
| Hardness, Rockwell C       | 38                       | 38                         |          |
| Tensile Strength, Ultimate | <a href="#">1170</a> MPa | <a href="#">170000</a> psi |          |
| Tensile Strength, Yield    | <a href="#">965</a> MPa  | <a href="#">140000</a> psi |          |
| Elongation at Break        | 6.0 %                    | 6.0 %                      |          |
| Reduction of Area          | 25 %                     | 25 %                       |          |
| Modulus of Elasticity      | <a href="#">204</a> GPa  | <a href="#">29600</a> ksi  |          |

| Electrical Properties  | Metric                           | English                          | Comments |
|------------------------|----------------------------------|----------------------------------|----------|
| Electrical Resistivity | <a href="#">0.0000830</a> ohm-cm | <a href="#">0.0000830</a> ohm-cm |          |

| Thermal Properties                                                                                       | Metric                                                      | English                                                                  | Comments |
|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------|----------|
| CTE, linear           | <a href="#">11.0</a> µm/m-°C<br>@Temperature 0.000 - 100 °C | <a href="#">6.11</a> µin/in-°F<br>@Temperature 32.0 - 212 °F             |          |
|                                                                                                          | <a href="#">11.6</a> µm/m-°C<br>@Temperature 0.000 - 315 °C | <a href="#">6.44</a> µin/in-°F<br>@Temperature 32.0 - 599 °F             |          |
| Specific Heat Capacity                                                                                   | <a href="#">0.460</a> J/g-°C<br>@Temperature 0.000 - 100 °C | <a href="#">0.110</a> BTU/lb-°F<br>@Temperature 32.0 - 212 °F            |          |
| Thermal Conductivity  | <a href="#">16.4</a> W/m-K<br>@Temperature 100 °C           | <a href="#">114</a> BTU-in/hr-ft <sup>2</sup> -°F<br>@Temperature 212 °F |          |
|                                                                                                          | <a href="#">21.8</a> W/m-K<br>@Temperature 500 °C           | <a href="#">151</a> BTU-in/hr-ft <sup>2</sup> -°F<br>@Temperature 932 °F |          |
| Melting Point                                                                                            | <a href="#">1400 - 1450</a> °C                              | <a href="#">2550 - 2640</a> °F                                           |          |
| Solidus                                                                                                  | <a href="#">1400</a> °C                                     | <a href="#">2550</a> °F                                                  |          |
| Liquidus                                                                                                 | <a href="#">1450</a> °C                                     | <a href="#">2640</a> °F                                                  |          |

| Component Elements Properties | Metric        | English       | Comments   |
|-------------------------------|---------------|---------------|------------|
| Aluminum, Al                  | 0.75 - 1.5 %  | 0.75 - 1.5 %  |            |
| Carbon, C                     | <= 0.090 %    | <= 0.090 %    |            |
| Chromium, Cr                  | 16 - 18 %     | 16 - 18 %     |            |
| Iron, Fe                      | 70.6 - 76.8 % | 70.6 - 76.8 % | as balance |
| Manganese, Mn                 | <= 1.0 %      | <= 1.0 %      |            |
| Nickel, Ni                    | 6.5 - 7.75 %  | 6.5 - 7.75 %  |            |
| Phosphorus, P                 | <= 0.040 %    | <= 0.040 %    |            |
| Silicon, Si                   | <= 1.0 %      | <= 1.0 %      |            |
| Sulfur, S                     | <= 0.040 %    | <= 0.040 %    |            |

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.

